

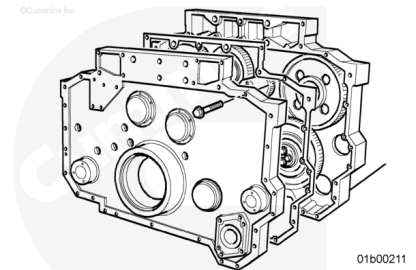
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>)



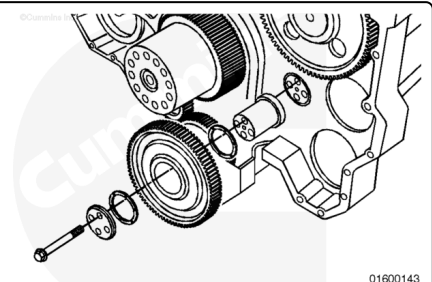
Remove

Mark the flange to show oil hole orientation.

Remove the four capscrews, the thrust retainer plate, and the thrust bearings on the left-bank hydraulic pump drive idler gear.

Remove the hydraulic pump drive idler gear.

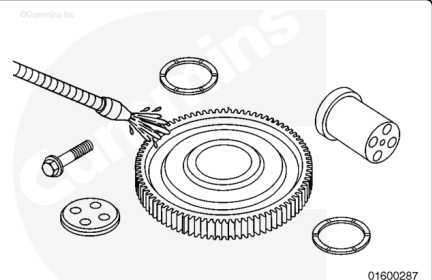
Remove the idler shaft.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



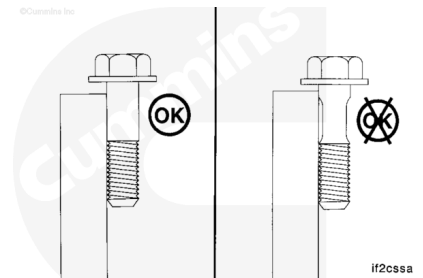
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the parts with solvent and dry with compressed air.

Use a straightedge to inspect the retaining capscrews for necking.

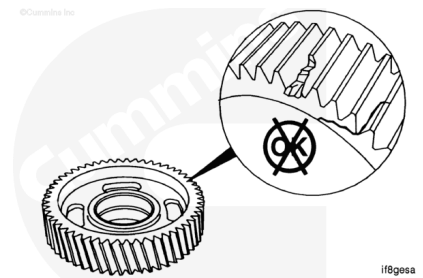
If the retaining capscrews are necked, they **must** be replaced.



Inspect the idler gear for chipped, broken, or cracked teeth.

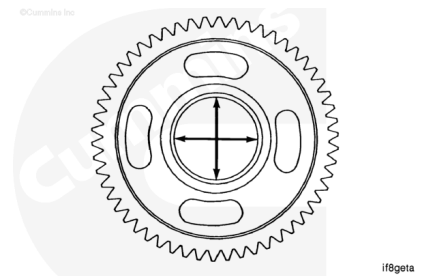
Inspect the bushing for damage.

If the idler gear bushing or teeth are damaged, the idler gear **must** be replaced.



Measure the inside diameter of the idler gear bushing.

Hydraulic Pump Drive Idler Gear Inside Diameter		
mm		in
73.035	MIN	2.8753
73.065	MAX	2.8765



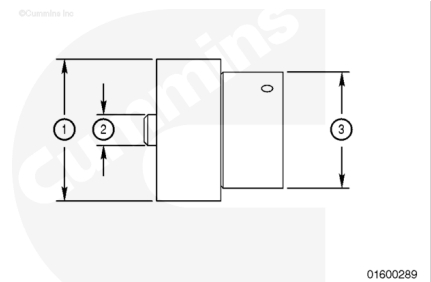
The bushing in the idler gear is precision bored after installation. Replace the bushing and gear as an assembly if the idler gear bushing is **not** within specification.

If the idler gear is a compound gear it **must** be replaced as an assembly.

Measure the outside diameter of the idler gear shaft.

Hydraulic Pump Drive Idler Gear Shaft Outside Diameter			
	mm		in

(1)	88.75	MIN	3.494
	89.05	MAX	3.506
(2)	21.975	MIN	0.8652
	22.000	MAX	0.8661
(3)	72.92	MIN	2.871
	72.98	MAX	2.873

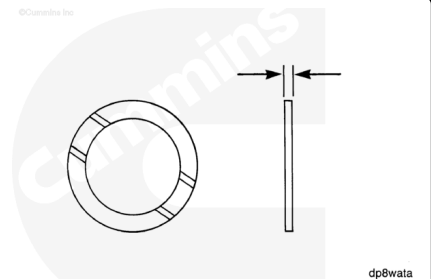


If the idler gear shaft is **not** within specifications, it **must** be replaced.

Check the grooved side of the thrust bearing for damage.

If the grooved side of the thrust bearing is damaged, it **must** be replaced.

Measure the thrust bearing thickness.



Hydraulic Pump Idler Gear Thrust Bearing Thickness

mm		in
2.337	MIN	0.092
2.387	MAX	0.094

If the thrust bearing is **not** within specification. The thrust bearing **must** be replaced.

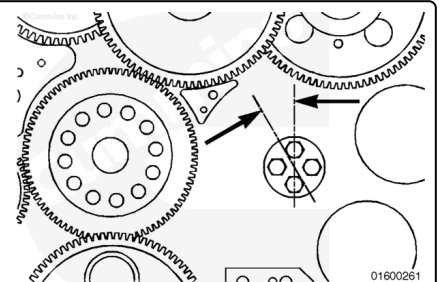
Install

⚠ CAUTION ⚠

To reduce wear on the idler shaft, it must be installed in the correct position.

Install a guide stud into the uppermost mounting hole location for the idler shaft.

Install the idler shaft over the guide stud with the oil hole in the 11 o'clock position and mate the shaft to the cylinder block face.



Install the thrust washer over the idler shaft, the grooved side **must** point outward from the block.

Lightly oil the face of the thrust washer with clean engine oil.

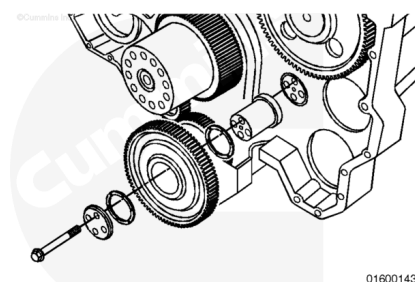
Lubricate the gear bushing with clean engine oil.

Install the drive gear onto the idler shaft with the small diameter gear towards the cylinder block.

Install the second thrust washer with the grooved side toward the engine.

Lubricate the thrust washer with clean engine oil.

Install the idler shaft retaining washer over the guide stud and line up with the mounting holes.



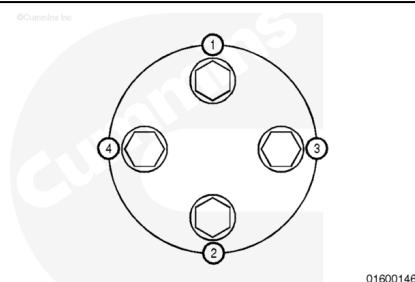
Install two capscrews.

Remove the guide stud and install the remaining two capscrews.

Tighten the capscrews in the sequence shown.

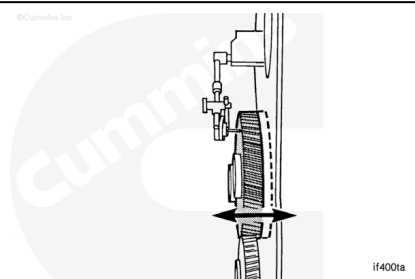
Torque Value:

1. 80 n•m [59 ft-lb]
2. 165 n•m [122 ft-lb]
3. 280 n•m [207 ft-lb]



Use a dial indicator to measure the idler gear end clearance.

Hydraulic Pump Idler Gear - End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013



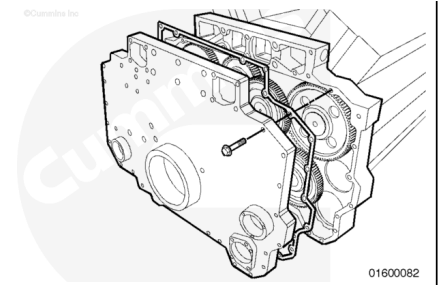
If the clearance is **not** within specifications, check for foreign material between the parts, or check for proper location of the thrust washer. Oversize thrust washers are available.

Finishing Steps

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>)
- Operate the engine and check for leaks.



Last Modified: 23-Aug-2018
